SYSTEM ANALYSIS DOCUMENTATION (SAD)

UMHackathon 2026

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1. Introduction:

This document introduces AdPilot, a web-based AI-powered advertising decision and content generation system built to solve the problem of SME e-commerce sellers lacking structured, data-driven advertising guidance. AdPilot addresses this by leveraging Z.AI's GLM as its sole intelligence engine to transform a seller's uploaded inventory data into actionable advertising strategies and ready-to-publish ad content.

1.1. Purpose:

The key element of system that has been covered are as followed:

1. **Architecture**:
    a. Frontend Layer: Built using React to provide the user interface and handle user interactions.

    b. API/Backend Layer: Uses Python, FastAPI, and Uvicorn to manage API requests and coordinate communication between frontend and backend services.
    c. Risk Analysis Module: A Python-based service that validates incoming requests and ensures safe processing.
    d. Live Context Gathering: Powered by GLM-5.1 to collect relevant real-time context for accurate responses.
    e. Strategy Decision Module: Uses GLM-5.1 to determine the best response strategy based on user input and gathered context.
    f. Content Generation Module: Uses GLM-5.1 to generate text-based outputs.
    g. Image Generation Module: Uses GLM-Image to generate visual content when needed.
    h. Explanation Module: Powered by GLM-5.1 to provide reasoning and explanations for generated outputs.
    i. Database Layer: Uses Firestore to store user data, generated content, logs, and contextual information.
2. **Data Flows:**
    a. User uploads a CSV inventory file into the system. The uploaded data is then processed through the FastAPI backend, where Z.ai's GLM analyzes the seller's inventory data and executes the four-stage agentic pipeline to generate advertising recommendations and content outputs.
3. **Model Process:**
    a. The end-to-end workflow begins with the seller uploading a CSV inventory file, followed by GLM-powered inventory classification, advertising strategy generation, seller review and selection, ad content generation, and final content export.
4. **Role of Reference:**

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a. This document serves as the technical reference for all developers and testers to understand AdPilot's architecture, data flows, and system boundaries throughout UMHackathon 2026.

1.2. Background:

E-commerce sellers in Southeast Asia, particularly SMEs operating on Shopee, Lazada, and TikTok Shop, consistently struggle with advertising decisions. Most rely on guesswork rather than data, wasting budget on poorly timed and poorly targeted campaigns. Existing platform tools such as Shopee Ads and TikTok for Business help sellers execute ads but do not tell them what to advertise, where, when, or to whom. AdPilot was built to fill this strategy gap using Z.AI's GLM as the decision-making engine.

1. **Previous Version:**
   a. AdPilot Version 1.0 is the initial release developed for UMHackathon 2026. There is no prior version. The system is built from the ground up as a hackathon project.

2. **New Capabilities Introduced**

   a. Inventory-to-strategy pipeline powered by Z.AI's GLM: Classifies uploaded seller inventory and generates ranked advertising strategy recommendations per session
   b. Four-stage agentic GLM prompting pipeline with output validation at each reasoning stage
   c. Seasonal and trend-aware recommendation engine that aligns ad strategies with upcoming e-commerce sales events
   d. Ad content generation platform-specific copy, captions, hashtags, and posting guidance generated from the seller's selected strategy
   e. CSV upload caching: User inventory uploads and generated outputs are stored as session history to avoid redundant GLM calls on repeat submissions
   f. Personal account system: Sellers can register, log in, and access their recommendation and content history

1.3. Target Stakeholder

Stakeholders	Roles	Expectations
SME Sellers	Upload inventory CSV and receive ranked advertising strategy recommendations	Faster, data-driven advertising decisions

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	and ready-to-publish ad content	without requiring marketing expertise
Marketing Agencies	Manage and content across multiple seller client's campaigns	Reduce manual workload and faster campaign turnaround for clients
Social Commerce Creators	Generate platform-native captions, hashtags, and posting guidance for TikTok and Instagram	Faster content production aligned with current platform trends
Development Team	Build and maintain the React frontend, FastAPI backend, GLM service layer, and SQL database	Modular and well-structured architecture that is clear to work with
QA Team	Test pipeline reliability, GLM response validation, and end-to-end flow accuracy	Clear workflows, defined input and output schemas, and reproducible test scenarios

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2. System Architecture & Design

2.1. High Level Architecture

2.1.1. Overview:

Type	Details
System	Browser-based web application accessible on desktop and mobile devices
Architecture	Cloud-server architecture with modular microservices
Frontend	React
Backend	Python
API Framework	FastAPI
Server	Uvicorn
Database	Firebase Firestore
AI Service	ILMU – GLM-5.1 (Text Processing) Z.AI – GLM-Image (Image Generation)
Deployment	Vercel (Frontend) AWS (Backend)

AdPilot uses a client-server web architecture. The React frontend handles user interactions such as login, registration, CSV uploads, dashboard display, ad previews, and exports.

The backend, built with Python, FastAPI, and Uvicorn, manages API requests, CSV validation, AI workflow orchestration, session caching, and data storage in Firebase Firestore.

For AI services, AdPilot integrates ILMU's GLM-5.1 for text reasoning and content generation, and Z.AI's GLM-Image for image generation. Firebase also supports authentication and real-time data management.

2.1.2 LLM as Service Layer

GLM is integrated into AdPilot as the structured intelligence backbone of the entire pipeline. The following describes how GLM operates at these stage:

- **Stage 1 - Inventory Classification:**

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The FastAPI backend constructs a prompt containing the parsed and validated CSV product data. GLM is instructed to classify each product by promotional urgency and seasonal relevance. The response is returned as structured JSON and validated against a schema.

• **Stage 2 - Strategy Recommendation Generation:**
The backend constructs a new prompt containing the inventory classification output plus injected context about current SEA market trends and the seasonal event calendar. GLM generates at least 2 ranked advertising strategy recommendations covering platform, ad format, pricing strategy, target audience, and best posting time with plain-language reasoning. The response is returned as structured JSON and validated before being stored and returned to the frontend.

• **Stage 3 - Ad Content Generation:**
The backend constructs a prompt containing the selected strategy and the relevant product details from the CSV. GLM generates the complete ad content package including headline, body copy, platform-optimised caption, hashtag set, and posting guidance. A post-generation validator checks that the content accurately reflects the product name, category, and pricing from the seller's input before it is shown.

2.1.3 Dependency Diagram

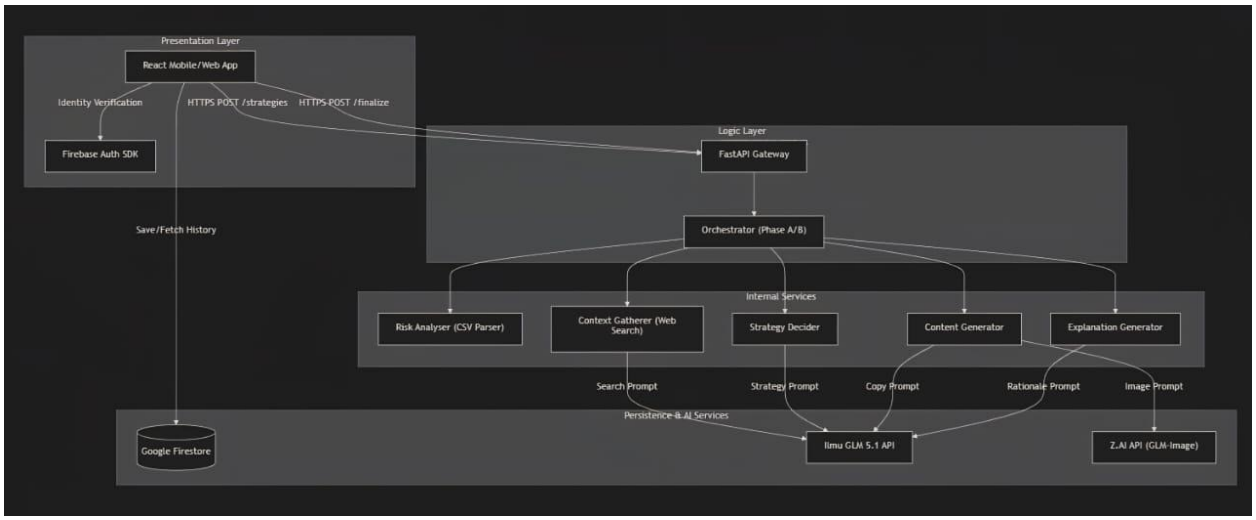

Figure 1 Dependency Diagram

2.1.4 Sequence Diagram

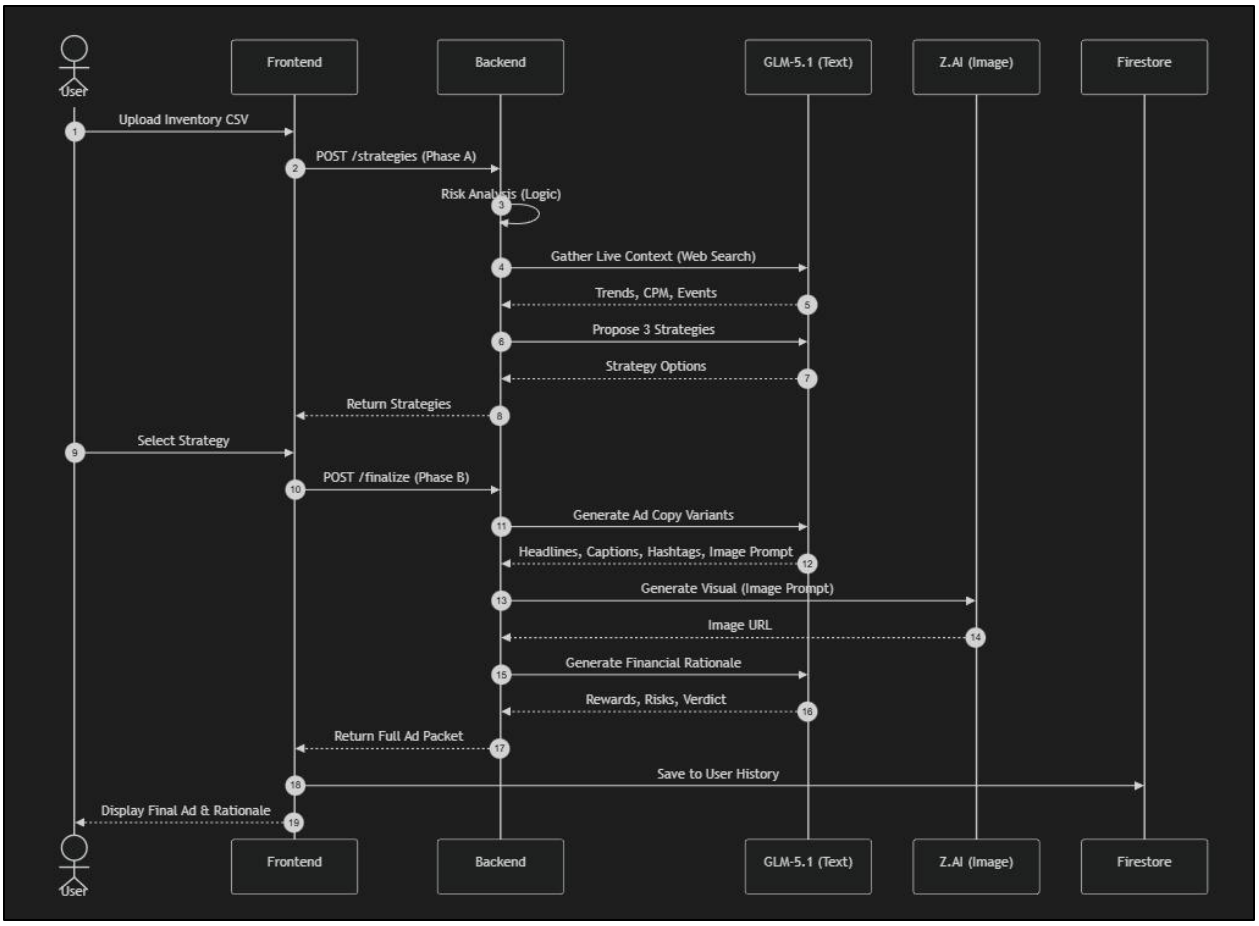

Figure 2 Sequence Diagram

2.2. Technological Stack

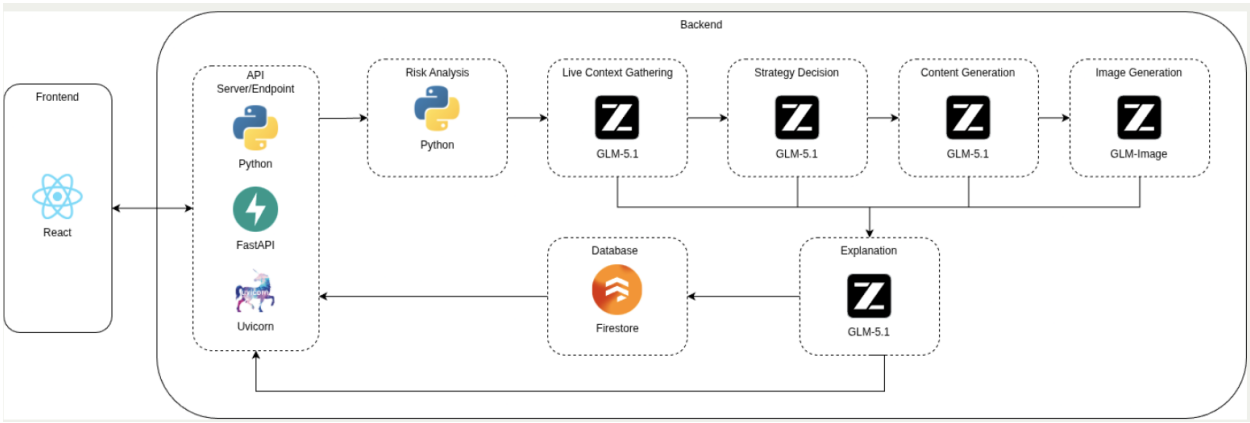

Figure 3 Technological Stack

Key Data Flow

2.2.1. Data Flow Diagram (DFD)

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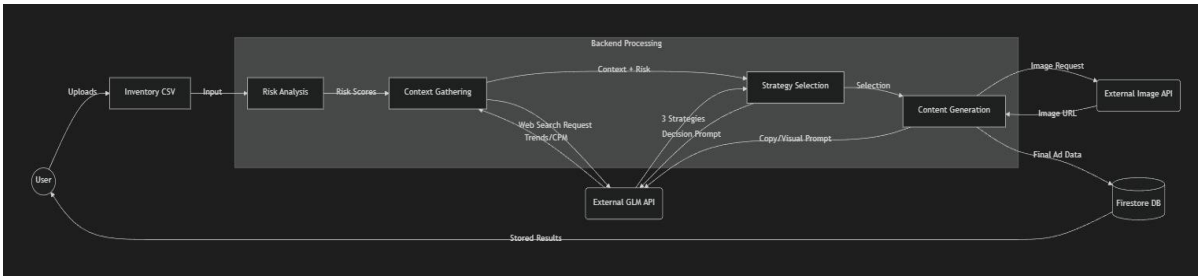

Figure 4 Data Flow Diagram

2.2.2. Normalized Database Schema

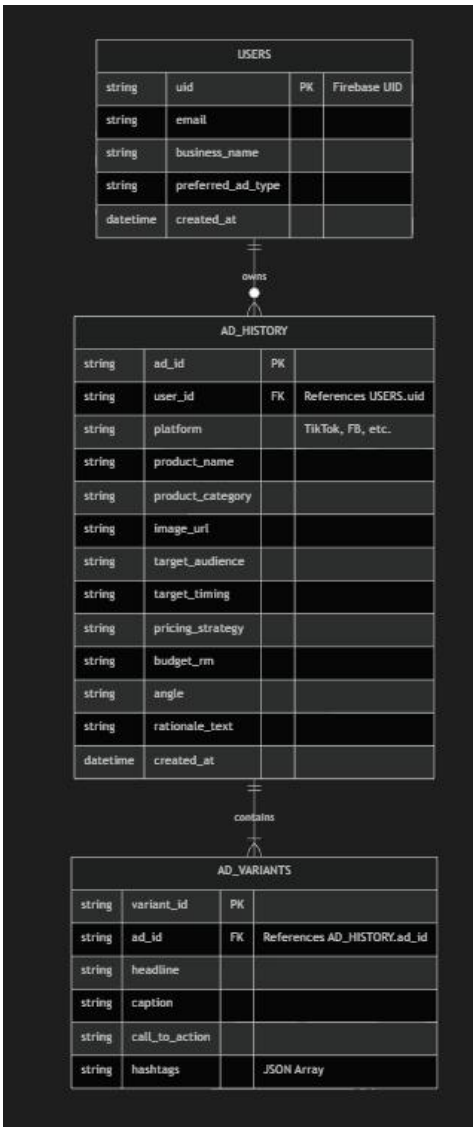

Figure 5 Normalize Database Schema

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3. Functional Requirements & Scope

3.1. Minimum Viable Product:

3.1.1. The following three core features will be built and demonstrated by the end of UMHackathon 2026:

No.	Features	Description
1	Inventory Analysis	Analyzes the uploaded inventory to identify overstocked products, new products, and seasonal opportunities based on upcoming Southeast Asian shopping events.
2	Advertising Strategy Recommendations	After analyst the store, the system generates 3–5 personalized advertising recommendations based on inventory data, market trends, seasonal relevance, platform selection, pricing strategy, target audience, and posting time.
3	Ad Content Generation	Based on the recommendation strategy, GLM generates ready-to-publish ad copy, captions, hashtags, and posting guidance for platforms like Shopee, TikTok Shop, Instagram, and Facebook.

3.2. Non-Functional Requirements (NFRs)

Quality (e.g.)	Requirements (e.g.)	Implementation (e.g)
Scalability	AdPilot must handle increased concurrent user sessions during peak e-commerce event periods such as 11.11 and Hari Raya	FastAPI's async architecture handles concurrent requests efficiently. CSV upload caching reduces live GLM

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	without significant degradation in response time.	API calls during peak periods by serving repeated inventory analyses from session history.
Reliability	The recommendation pipeline must return a valid output or a clear error state. No hallucinated or schema-invalid content should reach the seller interface under any circumstance.	JSON schema validation is applied to all Stage 1 and Stage 2 GLM responses, with up to two automatic retries per stage on failure. A post-generation content check is applied at Stage 4. A plain-language error message is shown to the seller if all retries fail.
Maintainability	The CSV parser, GLM pipeline stages, and database operations should be independently modifiable without breaking other system components.	Each pipeline stage is implemented as an isolated function in the backend with clearly defined input and output contracts. The SQL schema is normalised to 3NF to support clean data operations.
Token Latency	Stage 1 and Stage 2 GLM calls should return within 15 seconds under normal load. Stage 4 should return within 10 seconds.	FastAPI async endpoints handle GLM calls without blocking. The frontend displays a loading indicator while awaiting responses. Chunking is applied for large inventories to keep per-call

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		token size manageable and latency predictable.
Cost Efficiency	Average token cost per user session must remain within a sustainable threshold relative to the subscription pricing model.	CSV upload caching eliminates redundant GLM calls for repeat inventory submissions. Structured JSON output schema reduces output token consumption. Files exceeding 500 rows are rejected before any API call is made to prevent unexpectedly large token costs.

3.3. Out of Scope / Future Enhancements

a. Direct integration with Shopee Ads, TikTok Ads Manager, Instagram, and Facebook for one-click campaign publishing

b. Real-time inventory synchronization with seller stores through API integrations instead of manual CSV uploads

c. AI-generated product visuals, banners, and short-form video ads for platforms such as TikTok and Instagram Reels

d. Campaign performance analytics dashboard to track conversions, ROI, click-through rates, and ad effectiveness

e. Autonomous campaign optimization that automatically adjusts budget allocation, targeting, and ad scheduling based on live campaign performance

f. Multilingual content generation for additional Southeast Asian languages beyond English

g. Agency dashboard for managing multiple seller accounts and campaigns simultaneously

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4. Monitor, Evaluation, Assumptions & Dependencies

4.1. Technical Evaluation:

4.1.1. Grayscale Rollout & A/B Testing:

New pipeline features such as an updated Stage 2 prompt strategy or a revised seasonal context injection approach are released to 5% of active users initially. System behaviour and recommendation quality are monitored over 24 to 48 hours before the rollout is incrementally expanded to 25%, 50%, and then 100% of users. Once a feature reaches full rollout, A/B testing is conducted where relevant. For example, comparing whether a few-shot Stage 2 prompt produces higher seller selection rates than a zero-shot variant before the better-performing version is fully adopted.

4.1.2. Emergency Rollback (ER) & Golden Release:

If a critical failure is detected such as a spike in GLM schema validation failures, hallucinated content confirmed to have reached the seller interface, or a sustained GLM API timeout rate exceeding the defined threshold, an emergency rollback is triggered. The deployment pipeline reverts to the last stable Golden Release of the backend service. Sellers in active sessions at the time of rollback receive a service interruption notice. Session state is preserved in the Firebase, so sellers can resume their session once the system is restored.

4.1.3. Priority Matrix:

Priority	Trigger Condition	Action
P1 - Critical	GLM schema validation failure rate exceeds 10% in any 10-minute window, or hallucinated content is confirmed to have reached the seller interface	Immediate emergency rollback to Golden Release. GLM pipeline suspended. Engineering team alerted. Sellers shown a service unavailability message.
P2 - High	Average Stage 1 or Stage 2 GLM response latency exceeds 20 seconds for more than 5% of requests in any 15-minute window	Backend async queue activated to process sessions in order and prevent API rate

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		limit breaches. Engineering team alerted to investigate root cause.
P3 - Medium	CSV upload caching miss rate is unexpectedly high, indicating more live GLM calls than projected and potential cost overrun	Session history matching logic reviewed. CSV hash comparison thresholds adjusted if necessary to improve cache hit accuracy.
P4 - Low	A seller reports a recommendation that is irrelevant to their inventory or ad content that misrepresents their product	Session log reviewed. Prompt engineering adjustment queued for next sprint. Seller offered a free session retry.

4.2. Monitoring

System Component	What is Monitored	Trigger Condition	Response Action
Backend Server (FastAPI)	Server uptime and API request failures	Server downtime or repeated failed requests	Restart service and notify developers
Frontend System	Response latency and failed AI requests	API timeout or repeated failures	Retry request and log issue
Z.ai GLM API	Database connectivity and authentication failures	Failed login or database errors	Alert developers and restore service
Firebase Database	Invalid file uploads and missing required fields	Upload validation failure	Display error message to users
CSV Upload System	High token consumption per session	Token usage exceeds threshold	Activate caching and usage limits

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Token Usage	UI crashes or export failures	Repeated frontend errors	Log issue and perform debugging

Assumptions:

a. Sellers have a stable internet connection when accessing the AdPilot web application and uploading their inventory CSV.

b. Sellers have access to their platform's to publish the content generated by AdPilot. AdPilot does not publish on the seller's behalf.

c. Z.AI's GLM API is available and performs within the defined latency parameters under normal operating conditions.

d. Sellers are assumed to have product inventory data that can be exported or converted into CSV format. The uploaded CSV file must contain five required fields: product_name, category, stock_level, price, and date_added to ensure accurate inventory analysis and recommendation generation.

e. The seasonal event calendar injected into Stage 2 prompts is maintained and updated before each major e-commerce sales event.

f. Firebase authentication services are operational and available during user login and registration.

4.3. External Dependencies:

Tools	Purpose	Risks
ILMU GLM 5.1	Core LLM reasoning engine — handles all prompt inference, multi-step reasoning, strategy recommendations, and ad content generation across the full pipeline.	High — API downtime or rate limits cause complete pipeline failure. Mitigation: async queue for concurrent sessions; graceful error state shown to seller if unavailable. graceful

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		degradation state from users]
Z.AI GLM Image	Generates ad image creatives as part of the final ad content package delivered to the seller.	High — image generation failure results in incomplete ad content. Mitigation: text content is returned first; seller is notified to retry image generation separately without restarting the full pipeline.
Firebase	Handles user authentication and serves as the primary database for storing user accounts, session history, uploaded CSV records, recommendation outputs, and generated content.	Medium — downtime prevents user login and blocks all data read/write operations. Mitigation: clear error message shown to user; session state preserved where possible for resumption once service is restored.

5. Project Management & Team Contributions

The duration of the Preliminary round is approximately 10 days considering the two-week sprint.

5.1. Project Timeline:

Project Name:	AdPilot
Project Start:	16/4/2026
Current Date:	26/4/2026
Weeks in Progress:	2

Task	Project Lead	Start Date	End Date	Days	Progress
Planning Stage					
Problem Research & Idea Validation	Chong Yun Enn	16/4/2026	16/4/2026	1	100%
Market Research & Competitor Analysis	Chong Yun Enn	16/4/2026	17/4/2026	2	100%
PRD Documentation	Chong Yun Enn	17/4/2026	18/4/2026	2	100%
System Architecture Design	Er Qing Yap	18/4/2026	19/4/2026	2	100%
UI Wireframe Design	Lim Tze Jiun	18/4/2026	19/4/2026	2	100%
Execution Stage					
Frontend Development	Lim Tze Jiun	19/4/2026	22/4/2026	4	100%
Backend Development	Er Qing Yap	19/4/2026	23/4/2026	5	100%
Database Setup	Lim Chun Rong	20/4/2026	21/4/2026	2	100%
GLM API Integration	Wong Yee Xiong	21/4/2026	23/4/2026	3	100%
Prompt Engineering & Testing	Wong Yee Xiong	22/4/2026	24/4/2026	3	100%
Recommendation Engine Integration	Er Qing Yap	23/4/2026	24/4/2026	2	100%
Content Generation Integration	Lim Chun Rong	23/4/2026	24/4/2026	2	100%
Review Stage					
System Testing	All Team Members	24/4/2026	25/4/2026	2	100%
Bug Fixing & Optimization	Lim Chun Rong	24/4/2026	25/4/2026	2	100%
Pitch Deck Preparation	Lim Tze Jiun	25/4/2026	25/4/2026	1	100%
Demo Video Recording	Wong Yee Xiong	26/4/2026	26/4/2026	1	100%
Final Presentation Rehearsal	All Team Members	26/4/2026	26/4/2026	1	100%

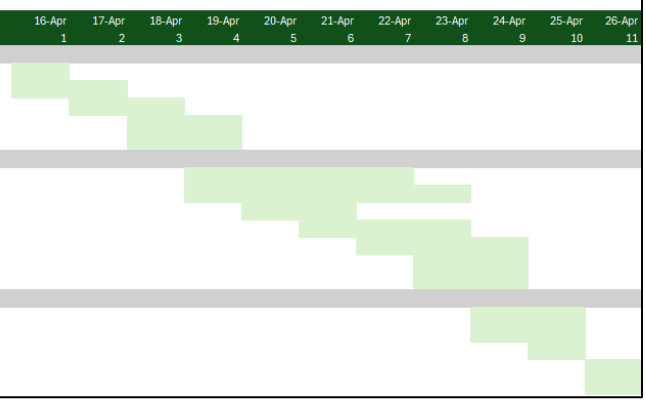

Figure 6 Project Timeline

Role of Team Members:

Name	Role
Chong Yun Enn	• Documentation Editor • Slide Editor • Video Editor
Lim Tze Jiun	• Documentation Editor • Slide Editor • Frontend Developer
Er Qing Yap	• Backend Developer
Lim Chun Rong	• Documentation Editor • Backend Developer
Wong Yee Xiong	• Documentation Edotor • Demo Producer • Backend Developer

5.2. Recommendations:

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- Implement the CSV session history cache using a hash of the file content (MD5 or SHA-256) as the cache key, so minor formatting differences between two otherwise identical uploads do not incorrectly bypass the cache.

- Use structured logging with a unique request ID per seller session to enable end-to-end tracing of each GLM pipeline call, making it straightforward to debug validation failures and retry events during the hackathon demo.

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